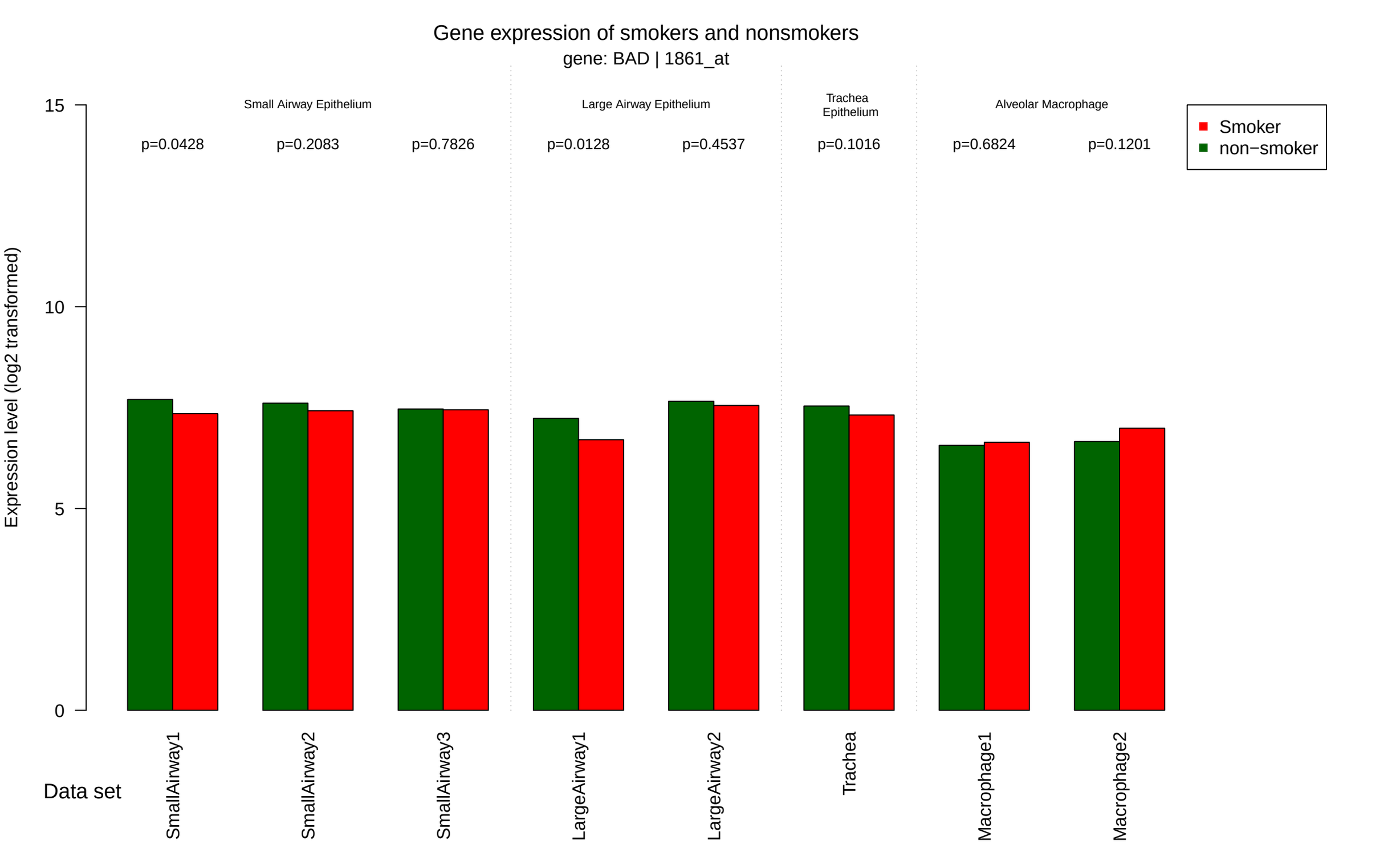
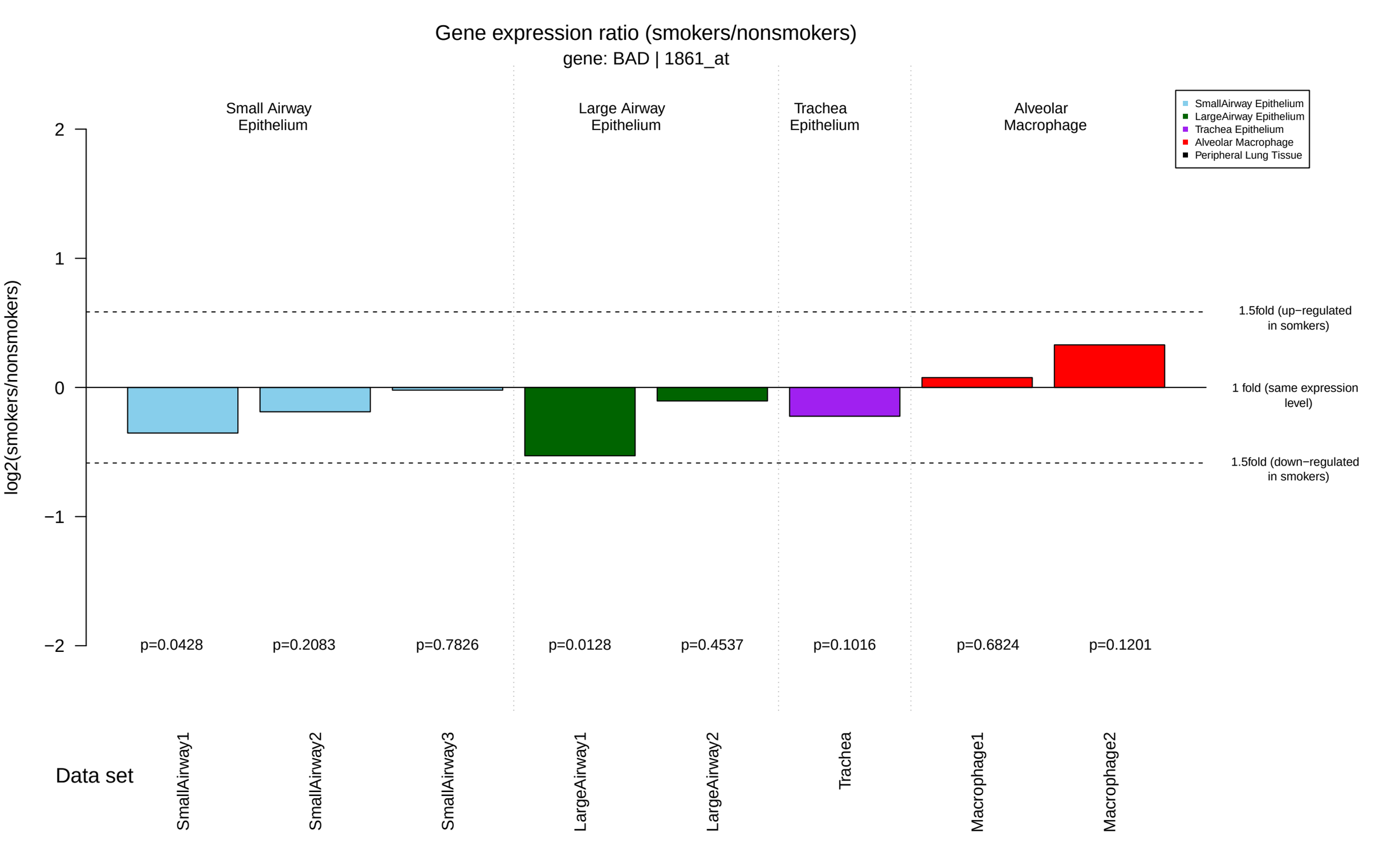
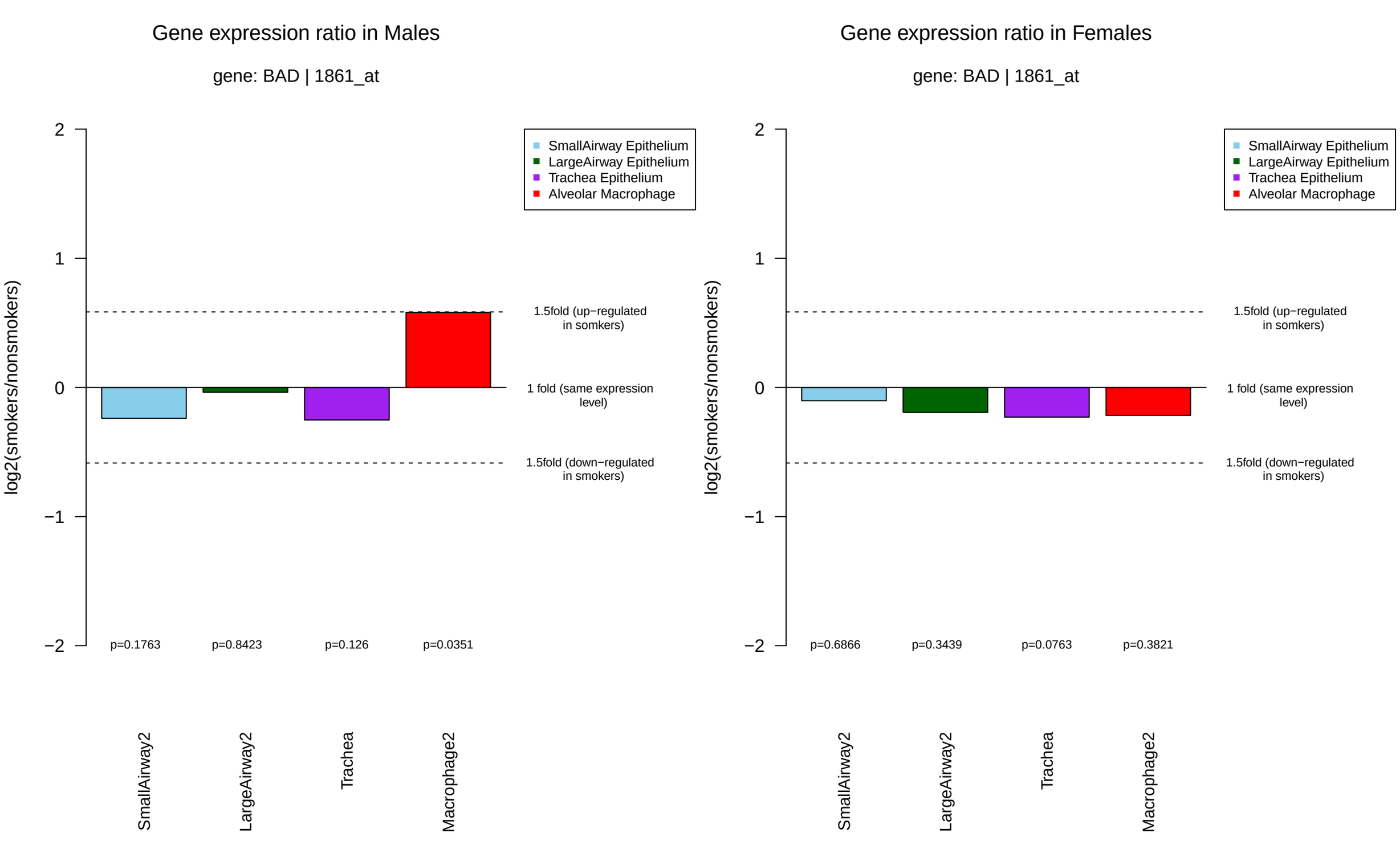
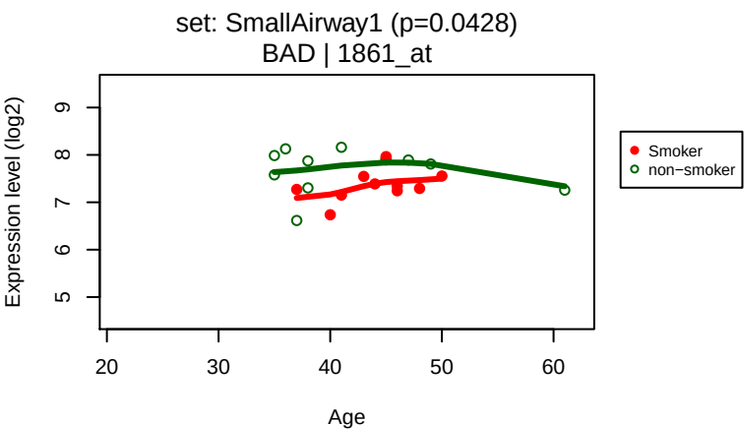
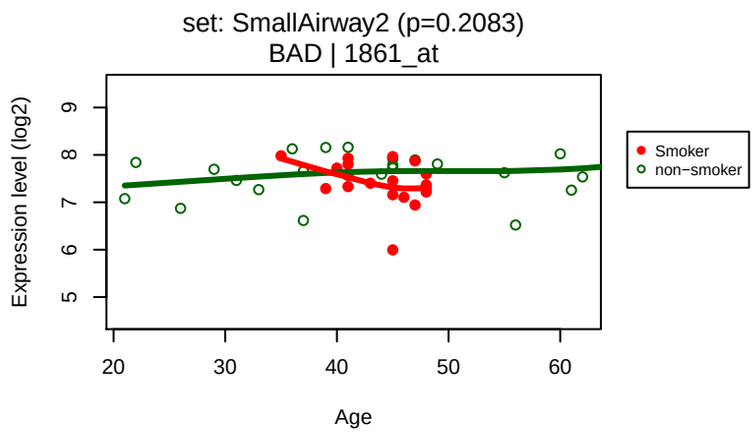
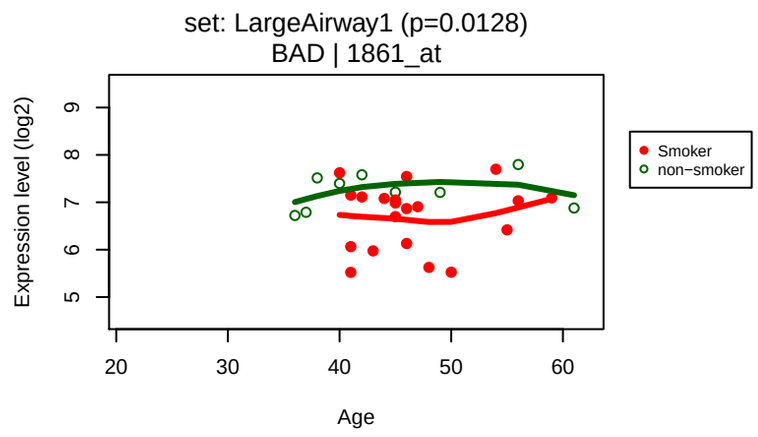

Gene expression of smokers and nonsmokers
gene: BAD | 1861_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0428
p=0.2083
p=0.7826
p=0.0128
p=0.4537
p=0.1016
p=0.6824
p=0.1201
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

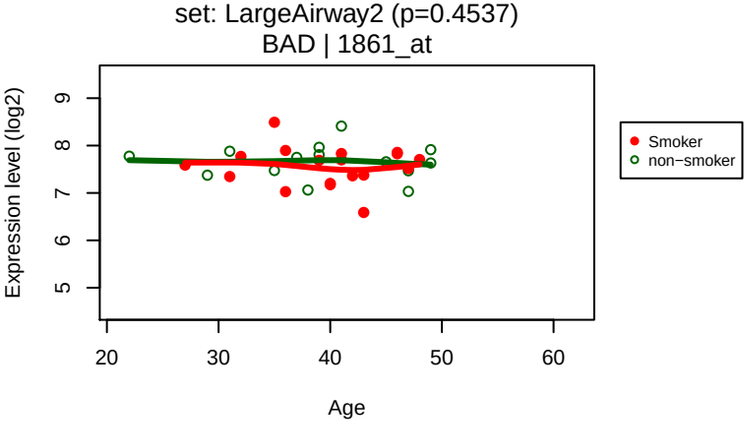

Gene expression ratio (smokers/nonsmokers)
gene: BAD | 1861_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0428
p=0.2083
p=0.7826
p=0.0128
p=0.4537
p=0.1016
p=0.6824
p=0.1201
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

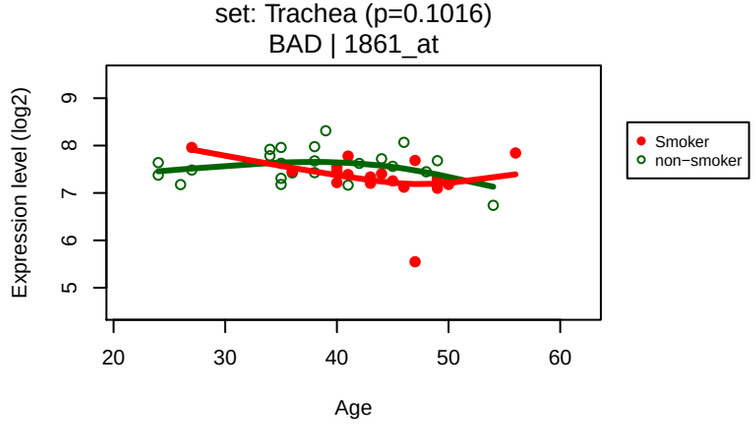

Gene expression ratio in Males
gene: BAD | 1861_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1763
p=0.8423
p=0.126
p=0.0351
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BAD | 1861_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6866
p=0.3439
p=0.0763
p=0.3821
SmallAirway2
LargeAirway2
Trachea
Macrophage2

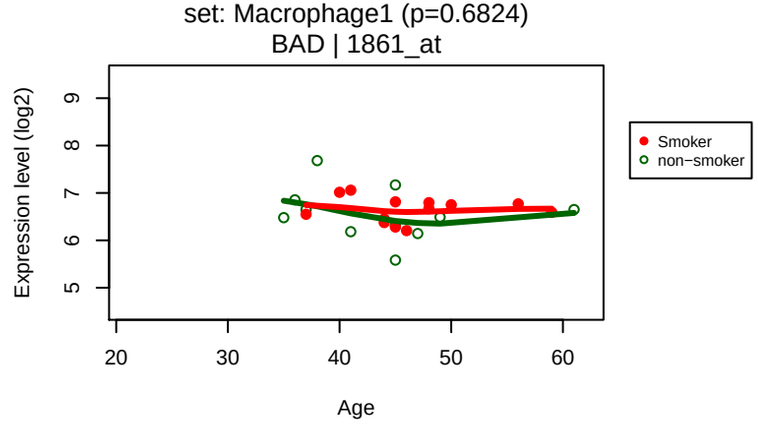

set: SmallAirway1 (p=0.0428)
BAD | 1861_at
Expression level (log2)
Age
Smoker
non−smoker

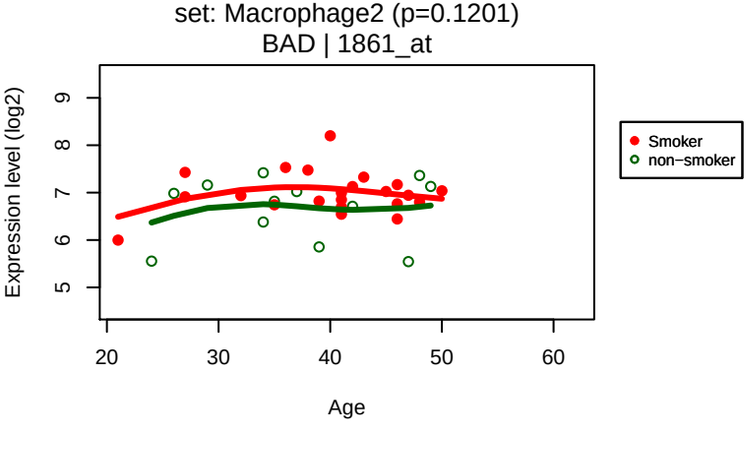

set: SmallAirway2 (p=0.2083)
BAD | 1861_at
Expression level (log2)
Age
Smoker
non−smoker

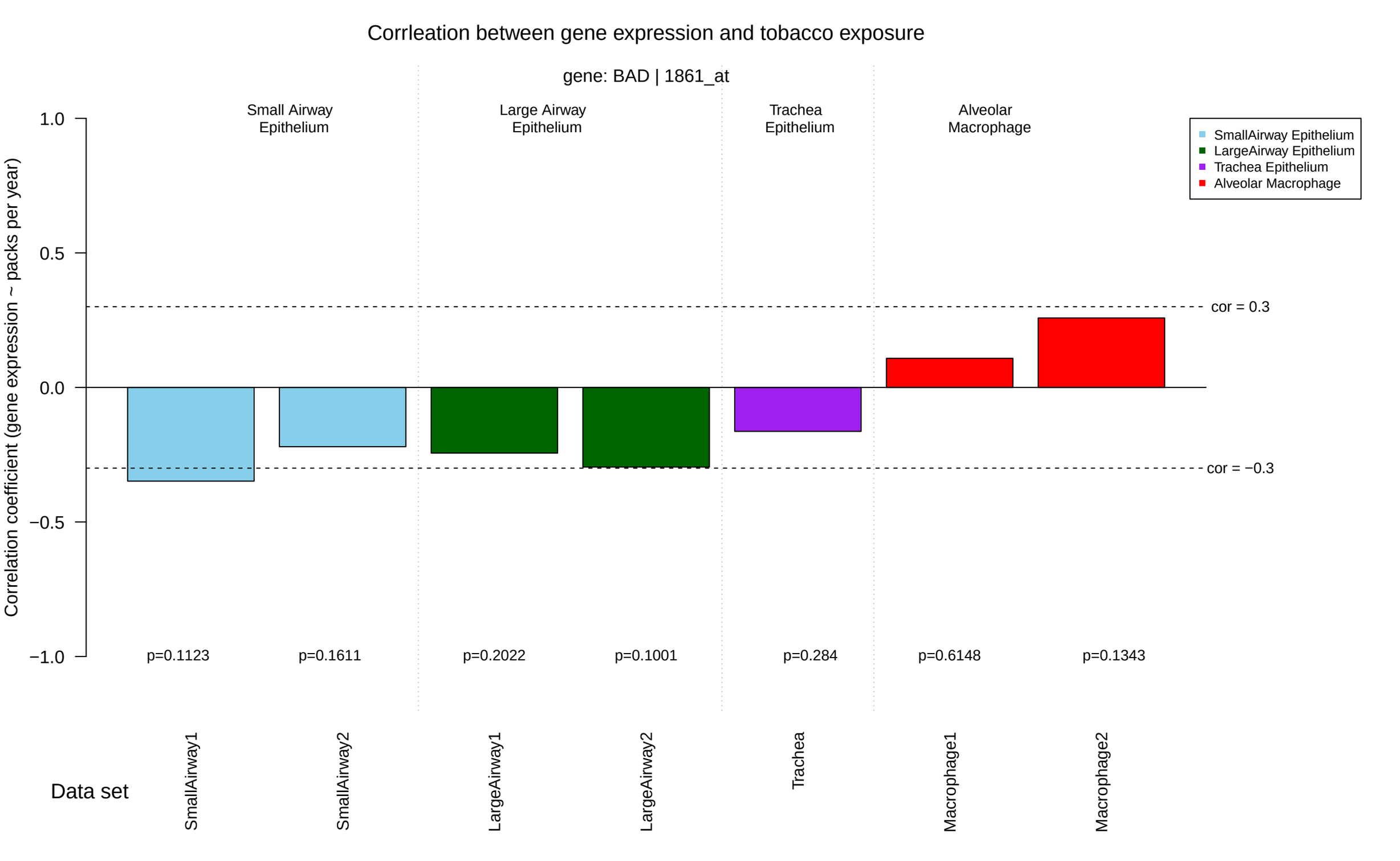

set: LargeAirway1 (p=0.0128)
BAD | 1861_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4537)
BAD | 1861_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1016)
BAD | 1861_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6824)
BAD | 1861_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1201)
BAD | 1861_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BAD | 1861_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1123
p=0.1611
p=0.2022
p=0.1001
p=0.284
p=0.6148
p=0.1343
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2